MIL-B-43159D
7 February 1979
SUPERSEDING
MIL-B-43159C
24 January 1972

MILITARY SPECIFICATION

BOX, TOTE, PLASTIC, HARDWARE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a plastic tote box of one type and size.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-1183	-	Plastic Molding Material, Acrylonitrile - Butadiene - Styrene (ABS), Rigid
UU-P-268	-	Paper, Kraft, Wrapping
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3990

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet Type Base

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

8-2-135 - Box, Plastic, Hardware

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard sample. The standard sample shall be used only in determining the stacking and nesting capabilities, as specified in 3.4 (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2 and 6.4).

3.3 Material (see 6.6).

3.3.1 Plastic. The plastic material used shall conform to type V of L-P-1183 or shall be high density linear polyethylene.

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3.4 Construction, design and dimensions. Construction, design and dimensions of the box shall be as shown on Drawing 8-2-135. As an alternate, the contractor may supply a box of his own design provided that the box conforms to all the requirements of this specification (including meeting and stacking properties), excepting design. The box may be made by thermoforming, vacuum forming, or any other suitable process. If the box is produced from sheet stock, the original sheet shall have a nominal initial thickness of 3/16 inch (5 mm). All boxes shall be constructed so that they will stack one on top of another and nest one inside another and stack and nest compatible with the standard sample (see 3.1).

3.4.1 Wall thickness. The wall thickness shall be not less than 0.075 inch (0.19 mm) except as follows:

(a) A minimum local wall thickness of 0.065 inch (0.17 mm) is permitted at the rounded corners of the stacking lugs.
(b) Thin spots not less than 0.040 inch (0.10 mm) are permitted within one inch (25 mm) of the stacking indentations.

3.5 Color. Unless otherwise specified (see 6.2), the color of the box shall be a neutral gray. When an exact color match is required, it shall be specified in the contract or order and a color sample supplied (see 6.3).

3.6 Finish. The interior surface of the box shall be smooth and have a luster finish as produced by molding or any other suitable commercial technique. Application of a finish or protective coating on interior or exterior surfaces shall not be allowed. The exterior surface finish of the box shall have a pebble or a simulated leather finish.

3.7 Performance.

3.7.1 Stacking and nesting. When the long axis of the box is parallel, the boxes shall nest together as shown on Drawing 8-2-135. When the long axis of the boxes are at right angles to each other, the boxes shall stack as shown on Drawing 8-2-135. The stacking lugs on one box shall fit firmly and securely into the stacking indentations on the corners of the box above. All boxes shall stack so that a stack of two or more boxes shall be firm and stable.

3.7.2 Physical characteristics.

3.7.2.1 Impact resistance room temperature. The finished box shall show no cracking, chipping or delamination when tested as specified in table II.

3.7.2.2 Impact resistance low temperature. The boxes shall show no cracking, chipping or delamination when tested as specified in table I.

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3.7.2.3 Heat resistance. The finished boxes shall show no distortion, delamination and no appreciable discoloration or loss of gloss, when tested as specified in table I. Appreciable means a change that is markedly noticeable when the tested sample box is compared with an untested sample box.

3.8 Identification marking. Each box shall be marked permanently and distinctly by a molded in process. The marking shall be located on the lip of the upper surface of the shorter axis of the box with the name or an easily identifiable trademark of the manufacturer.

3.9 Workmanship. The finished box shall be free from defects such as cracks, ripples, wrinkles, creases, pits and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required two sample boxes shall be submitted to determine compliance with 3.4. Any noncompliance with specified requirements shall be cause for rejection of the preproduction sample.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.2.5. The lot size shall be expressed in units of boxes for examination in 4.3.2.1 and 4.3.2.2, in units of shipping containers for examination in 4.3.2.3, and in units of palletized unit loads for examination in 4.3.2.4.

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4.3.2.1 Visual examination. The sample unit for this examination shall be one box. The box shall be examined for defects in color, finish, design, construction, workmanship and marking.

Examine	Defect
Quality and finish of plastic box	Color not as specified or does not match the approved sample. Interior surface not smooth, and not of uniform luster. Exterior surface not pebbled or not simulated leather. Any bubble, roughness, weld mark or shrink mark. Any cut, tear, hole, burn, break or mend in box. Any cracking, shipping, abrasion, distortion or delamination. Any wrinkle, ripple, crease, surface deterioration, foreign inclusion, or other imperfection.
Design and construction	Varies from specified design and construction or approved sample. Boxes do not stack as specified. Boxes do not nest as specified.
Workmanship	Mold flashing not removed. Any sharp or rough edge which may cause injury to users. Any dirt or other foreign matter on inside of box.
Identification marking	Missing, illegible, incorrect, imcomplete, not applied in specific location.

4.3.2.2 Dimensional examination. The sample unit for this examination shall be one box. Any dimension that is not within the specified tolerance shall be classified as a defect. To determine the wall thickness five determinations shall be made randomly over the entire area as applicable. The average of the five measurements shall be not less than the thickness specified in 3.4.1 or Drawing 8-2-135, as applicable. No local thickness shall be less than the minimum specified, when applicable.

4.3.2.3 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged except that it need not be closed. Examination for defects of closure shall be made on containers fully packaged.

Examine	Defect
Marking (exterior and unit packs)	Omitted; incorrect; incomplete; illegible, of improper size, location, sequence or method of application.
Materials	Any nonconforming components; component missing, damaged or otherwise defective.

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, improper taping or inadequate stapling. Bulged or distorted container.
Contents	Number per container is more or less than required.

4.3.2.4 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads (when applicable) not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.3.2.5 Inspection levels and acceptable quality levels (AQLs). The inspection levels for purposes of determining the sampel size, and acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2.1	I	4.0
4.3.2.2	S-2	6.5
4.3.2.3	S-2	4.0
4.3.2.4	S-1	6.5

4.3.3 Testing of the end item. The end item shall be tested for the applicable characteristics in table I. The sample unit shall be three finished plastic boxes. The lot size shall be the number of boxes in the end item inspection lot and shall be expressed in units of one box. The sample size shall be drawn at inspection level S-2 of MIL-STD-105 and the acceptable quality level (AQL) for each characteristic shall be 4.0 defects per one hundred units.

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INSTRUCTIONS FOR TESTING

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Impact resistance (room temperature)	3.7.2.1	4.4.1	X			X			
Impact resistance (low temperature)	3.7.2.2	4.4.2	X			X			
Heat resistance	3.7.2.3	4.4.3	X			X			

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4.4 Test methods.

4.4.1 Impact test. At room temperature 73° ± 3°F (23 ± 1°C) a box shall be turned upside down resting on the four stacking lugs. A steel ball weighing 4-pounds ± 2 ounces (1.81 ± 0.06 kg) shall be dropped from a height of 2-feet (0.6 m) so as to strike the outside of the center of the bottom of the box. The height shall be measured from the point of impact to the bottom of the ball. The ball shall be caught after the first bounce so that it strikes only once. The test shall be repeated three times and the surface of the box examined for cracks or breaks after each drop, both at the point of impact and on the reverse side (inside of the box).

4.4.2 Low temperature impact. The test in 4.4.1 shall be repeated with a new, i.e., not previously tested, box at 0° ± 2°F (-18° ± 1°C) and with the ball dropped from a height of 1-foot (0.3 m) instead of 2-feet (0.6 m). The box shall be cooled to 0° ± 2°F (-18° ± 1°C) temperature for not less than 1-hour prior to the test and also during the performance of the test.

4.4.3 Heat test. An untested box shall be exposed to 150° ± 5°F (66° ± 1°C) dry heat in an oven with forced air circulation for a period of 4-hours. After normal air cooling to room temperature 73° ± 3°F (23° + 1°C) the box shall be examined for defects including dimensional changes. Any change in dimensions which will adversely affect either the appearance, stacking, nesting or performance of the box shall be regarded as a defect. To determine stack and nesting characteristic requirements, all of the boxes shall be stacked and nested as specified in 3.7.1.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Six boxes shall be nested with a sheet of kraft paper completely and individually separating each box. The kraft paper shall overlap the outer edges of each box approximately 4 inches (102 mm). The kraft paper shall be of 30 pound minimum basis weight (49 g/m^2) conforming to type I, grade B of UU-P-268.

5.1.2 Commercial. Boxes shall be unit packed to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The unit pack and the quantity per unit pack shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Six boxes, unit packed as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed upright within a shipping container having approximate inside dimensions in inches of 25-1/4 x 19 x 16-1/2 (641 x 483 x 419 mm). Approximate inside dimensions are furnished as a guide only. The inside of each shipping container shall be provided with an A or C flute, top pad conforming to type CF, class domestic, variety SW, grade 275 of PPP-B-320. The top pad dimensions shall be 1/4 inch (6 mm) less than the inside length and width dimensions of the shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.3.

5.2.2 Level B packing. Six boxes, unit packed as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed upright within a shipping container having approximate inside dimensions in inches of 25-1/4 x 19 x 16-1/2 (641 x 483 x 419 mm). Approximate inside dimensions are furnished as a guide only. The inside of each shipping container shall be provided with an A or C flute, top pad conforming to type CF, class domestic, variety SW, grade 275 of PPP-F-320. The top pad dimensions shall be 1/4 inch (6 mm) less than the inside length and width dimensions of the shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.3.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.3.

5.2.3 Commercial packing. Boxes, unit packed as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), boxes packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with MIL-STD-147. Interlocking of loads (when applicable) shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The box covered by this specification is intended primarily to be used for handling and transporting small tools and mechanical or electrical parts. It is not intended for use with gasoline or with organic solvents, but may be used with water.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.2).
(c) Color required, if other than specified (see 3.5).
(d) Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).

6.3 Standard sample and color sample. For access to the standard sample and color sample, address the procuring office issuing the invitation for bids (see 3.5).

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one (1) unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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